Week 1:
Introduction and Computer Simulation

OUTLINE

- What is probability?
- Types of probability problems
- Probabilistic modeling
- Anaylsis versus computer simulation
- Why use computer simulation?
- Computer simulation of random experiments
- Determining characteristics of random variables
- Real-world example: digital communication

What is Probability?

- *Probability* is the chance that a given event will occur.
- Examples: the probability that it will rain tomorrow, the probability that you will win a lottery.
- Common thread of the examples given above is the presence of *a random experiment*, *a set of outcomes* and *probabilities* assigned to these outcomes.
- In the lottery case, for example, the experiment is the drawing, the outcomes are the lottery number sequences and the probabilities assigned are $1/N$ assuming that N is the total number of lottery number sequences.
- Another common thread is *statistical regularity* which means that averages obtained in long sequences of repetitions of random experiments consistently yield approximately the same value.

What is Probability?

- Average rainfall in the summer in Arizona from 1895 to 2002 is 4.40 inches.
- We can not predict the rainfall amounts for any given summer. However, probability theory enables us to predict the behavior of random experiments in the long run.

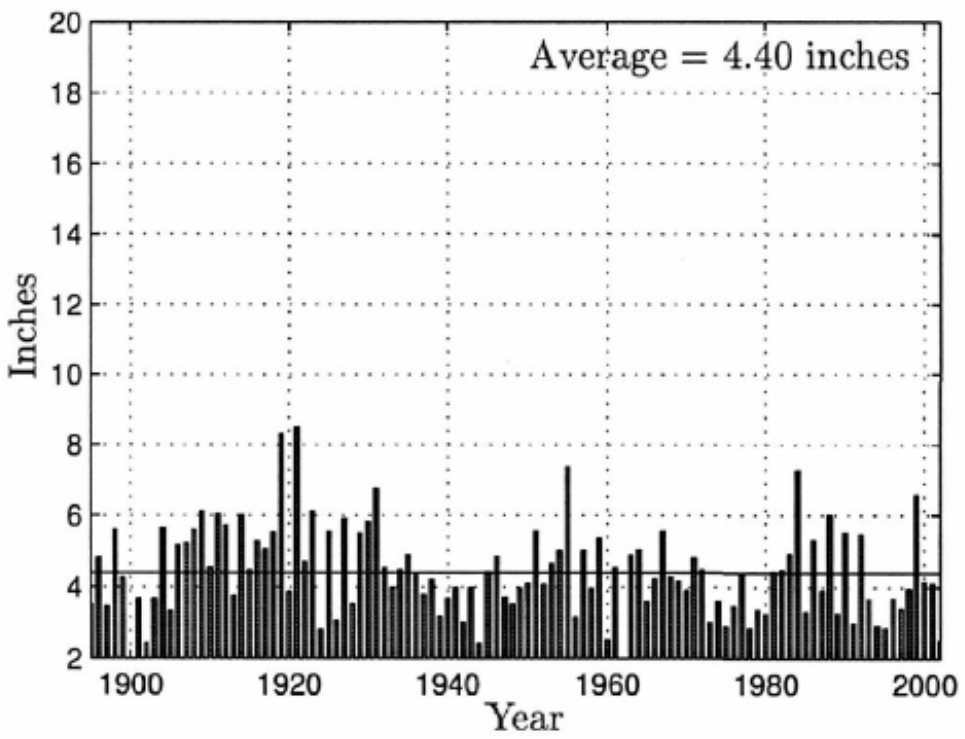


Figure source: S. M. Kay, *Intuitive Probabilty and Random Processes Using MATLAB*, Springer, 2006.

Types of Probability Problems

- Probabilistic methods are divided into two distinct types (*discrete* and *continuous*) based on the mathematics required to determine probabilities.

- Discrete approach is used when the number of outcomes is finite or infinite but countable. Otherwise, continuous model must be used.

Example: Suppose that there are N persons in an office. Consider the number of persons that are talking on the phone anytime between 9:00 AM and 9:10 AM. Possible outcomes are 0,1,2,…, N (discrete). If we are interested in the length of time a person is on the phone during that time period, the outcomes may be anywhere from 0 to 10 minutes (continuous).

In the first case, we can assign probabilities to the outcomes since the number of outcomes is finite. In the second case, it is not possible to assign a nonzero probability to each outcome as the outcomes are continuous.

Probabilistic Modeling

- Probability models are simplified mathematical approximations to reality.
- On the one hand, they should be comprehensive enough to represent characteristics of the random experiment, on the other hand they should be as simple as possible to be used in practice.
- Discrete or continuous probability models are used according to the underlying random experiment.
- In this course, we will assume that models are perfect and can be used to determine probabilities.
- In practice, the user must choose a reasonable model for a given application.

Probabilistic Modeling

Discrete Probability Model Example:
Suppose that there are *N* persons in an office. Propose a model for the probability of the number of persons being on the phone in the given 10-minute interval.

We can imagine a relationship between tossing a coin and the event of being on the phone (if heads appears, a person is on the phone, otherwise a person is not on the phone!). Then, the probability of the number of persons being on the phone is equal to the probability of the number heads in *N* coin tosses (in succesion or simultaneously).

We will see in the course that solution is the Binomial distribution. The probability of *k* heads out of *N* coin tosses is given by

$$\boldsymbol{P[k] = \binom{N}{k} p^k (1-p)^{N-k}, \qquad k = 0, 1, \ldots, N-1}$$

Probabilistic Modeling

The model must be modified to account for invalid assumptions, but it should exhibit the property of parsimony. Some objections to the model might be

- p, changes from person to person.
- Events are not independent.
- p changes over time.

$p = 0.75, N = 4.$

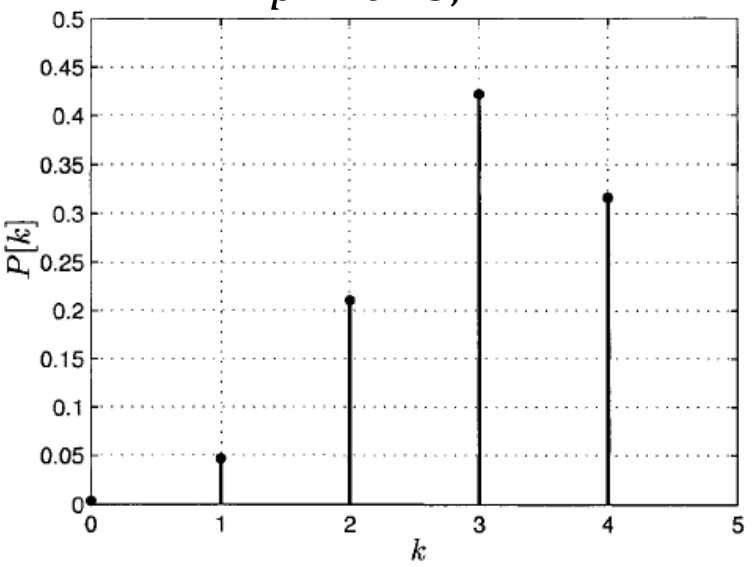


Figure source: S. M. Kay, *Intuitive Probabilty and Random Processes Using MATLAB*, Springer, 2006.

Probabilistic Modeling

Continuos Probability Model Example:
In the previous example, propose a model for the probability of the length of time T a person is on the phone in the given 10-minute interval.

In this case, the probability that T is equal to a number will be zero. Instead, we can talk about the probability that T is within an interval.

For continuous outcomes, Gaussian curve is frequently used as a probabilistic model. As we will see, for our problem the Gaussian model can be given by

$$P_T(t) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}(t-m)^2}, \qquad -\infty < t < \infty$$

Then, the probability that T is within an interval is obtained by integrating $p_T(t)$ over this interval.

Probabilistic Modeling

Since the area under the curve is zero if $T < 0$ or $T > 10$, the model is a reasonable approximation to physical reality. For example, the probability that a caller will be on the phone for between 5 and 6 minutes will be

$$P[5 \leq T \leq 6] = \int_5^6 p_T(t)dt = 0.1359$$

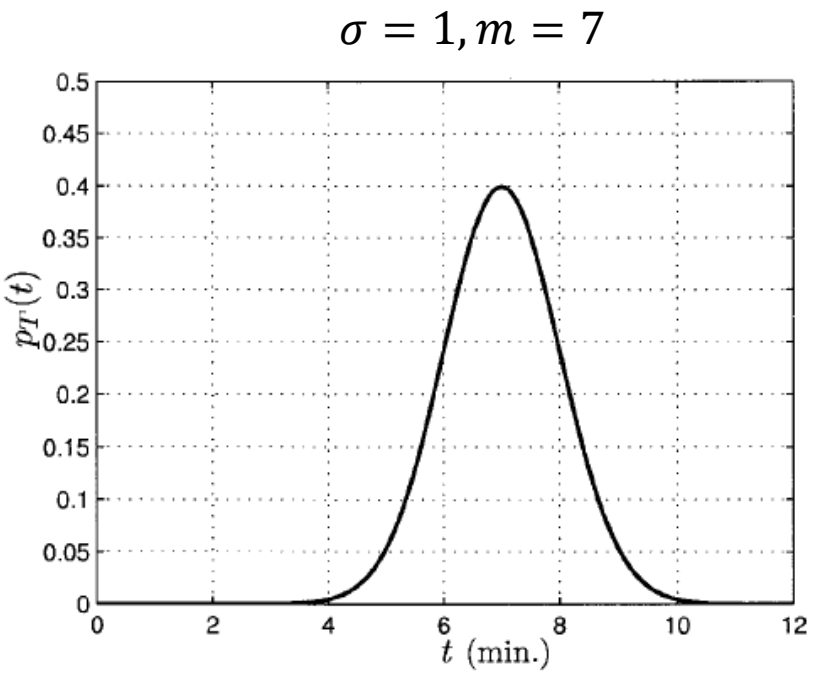


Figure source: S. M. Kay, *Intuitive Probability and Random Processes Using MATLAB*, Springer, 2006.

Analysis Versus Computer Simulation

- For many practical problems, it is not possible to determine probability functions.

- If we have a model for the random experiment, we can conduct the experiment a large number of times to obtain an approximate probability.

Example: Suggest a way to determine the probability of observing 3 heads in 4 tosses of a coin.

We toss the coin 4 times and count the number of heads (say x_1). We repeat the experiment yielding x_2. Continuing in this manner, we carry out 1000 repetitions to produce $(x_1, x_{2,} \ldots, x_{1000})$. Then, the probability of observing 3 heads can be approximated by

$$P[3 \text{ heads}] = \frac{\text{Number of times 3 heads observed}}{1000} = \frac{x_1 + x_2 + \cdots + x_{1000}}{1000}$$

Analysis Versus Computer Simulation

- This approach is tedious. Therefore, it is important to simulate this procedure.
- Today, we have strong computers carrying out a calculation 1,000,000 times in seconds. What is needed, then, is a way to simulate the toss of a coin.
- Most scientific software packages have built in *random number generators*. For example, in MATLAB a number in the interval (0,1) can be produced with the statement `x = rand(1, 1)`. The number is chosen at random. Hence, it is equally likely to be anywhere in the (0,1) interval.
- As a result, a number in the interval (0,1/2] will ve observed with probability 1/2. Similarly, a number in the interval (0,0.75] will be observed with probability 0.75.
- Any discrete random experiment can be simulated by using this approach.

Analysis Versus Computer Simulation

Example: Write a MATLAB code to simulate the number of heads in four tosses of a coin. Assume that probability of observing heads in a toss is $p = 0.5$.

```
counter = 0;
x = zeros(4);
for i = 1:4
if rand(1,1) < 0.5
  x(i) = 1;
  counter = counter + x(i);
end
```

Analysis Versus Computer Simulation

- For a continuous outcome experiment, a continium of outcomes must be generated.
- This is not possible since numbers are represented by a finite number of bits on digital computers.
- If the number of bits is large enough, approximation is adequate. For example, with 64 bits, 2^{64} numbers can be represented between 0 and 1 yielding a distance of $2^{-64} = 5 \times 10^{-20}$ between neighboring numbers.
- In MATLAB, numbers that follows a Gaussian curve can be generated by invoking the statement `randn(1, 1)`.

Why Use Computer Simulation?

- Computer simulations can be used for the following purposes:
  - providing counterexamples to proposed theorems
  - building intuition by experimenting with random numbers
  - lending evidence to a conjecture

- Consider the following question to illustrate the first case. Let U_1 and U_2 represent the outcomes of two independent experiments in which a number from 0 to 1 is chosen at random. What are the probabilities of $X = U_1 + U_2$?

- Let's assume that someone asserts that there is theorem that X is equally likely to be anywhere in the interval [0,2].

- We carry out a computer simulation by generating values of U_1 and U_2 and adding them together to see if the theorem is reasonable. We repeat this procedure M times and plot a histogram, which gives the number of outcomes that fall in each subinterval within [0,2].

Why Use Computer Simulation?

$M = 8$

Subintervals (bins): [0,0.5], (0.5,1], (1,1.5], (1.5,2].

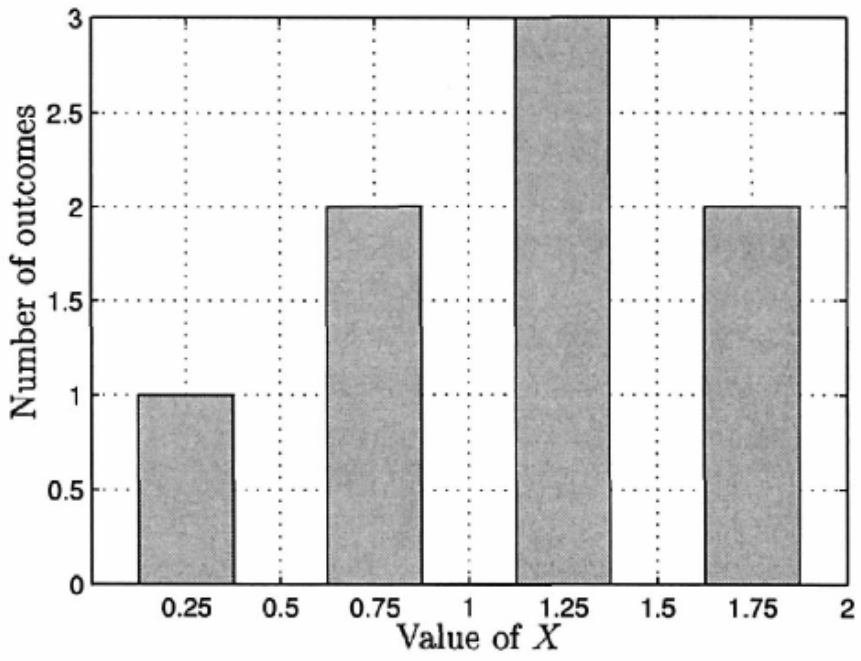
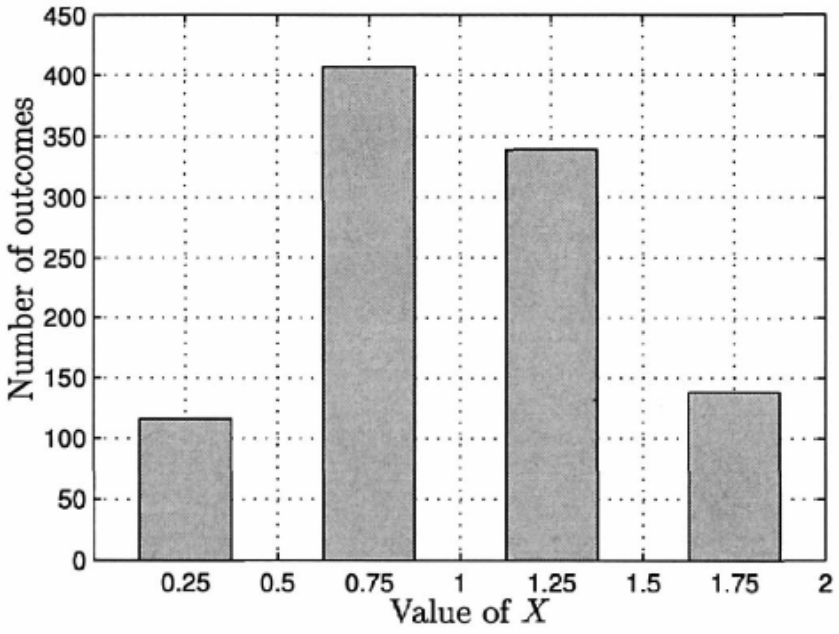


Figure source: S. M. Kay, *Intuitive Probability and Random Processes Using MATLAB*, Springer, 2006.

Why Use Computer Simulation?

$M = 1000$

bins: [0,0.5], (0.5,1], (1,1.5], (1.5,2].



It is clear that values of X are not equally likely. We have generated a counterexample to the proposed theorem.

Figure source: S. M. Kay, *Intuitive Probability and Random Processes Using MATLAB*, Springer, 2006.

Why Use Computer Simulation?

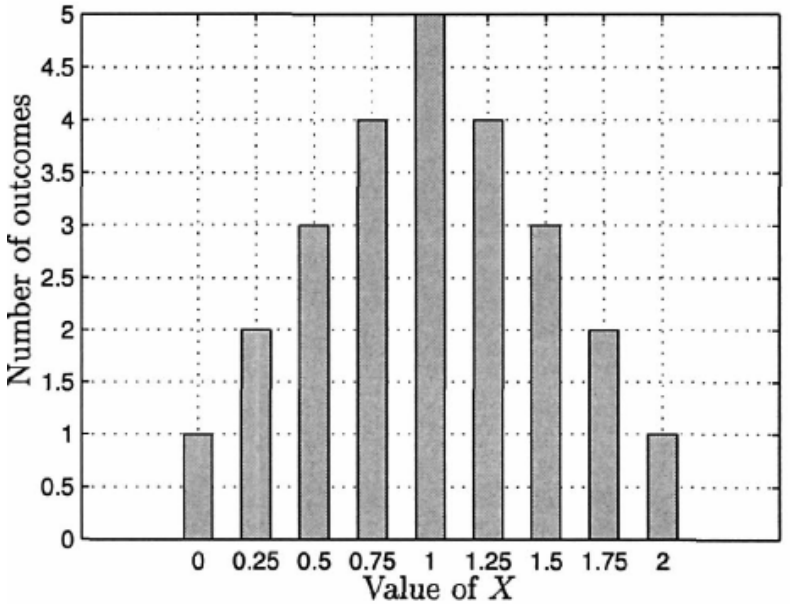
- Intuition Building: We might suppose from the histogram that probabilities of X are higher near one.
- This is reasonable because the number of combinations of U_1 and U_2 that will yield $X = 1$ is highest.
- To give an explanation, we look at a simpler problem in which U_1 and U_2 are assumed to take values in the set $\{0, 0.25, 0.5, 0.75, 1\}$.
- We repeat the experiment several times, find all values of $X = U_1 + U_2$ and plot the corresponding histogram.

Why Use Computer Simulation?

Possible values for $X = U_1 + U_2$

		U_2			
	0.00	0.25	0.50	0.75	1.00
0.00	0.00	0.25	0.50	0.75	1.00
0.25	0.25	0.50	0.75	1.00	1.25
U_1 0.50	0.50	0.75	1.00	1.25	1.50
0.75	0.75	1.00	1.25	1.50	1.75
1.00	1.00	1.25	1.50	1.75	2.00

Histogram for $X = U_1 + U_2, M = 25$



We have learned about what happens when outcomes of experiments are added together by employing computer simulation.

Figure and table sources: S. M. Kay, *Intuitive Probabilty and Random Processes Using MATLAB*, Springer, 2006.

Why Use Computer Simulation?

- Conjecture: As more and more experimental results are added together, a Gaussian shaped histogram is obtained.

- For lending credibility to this conjecture, we extend the previous experiment to three or more outcomes.

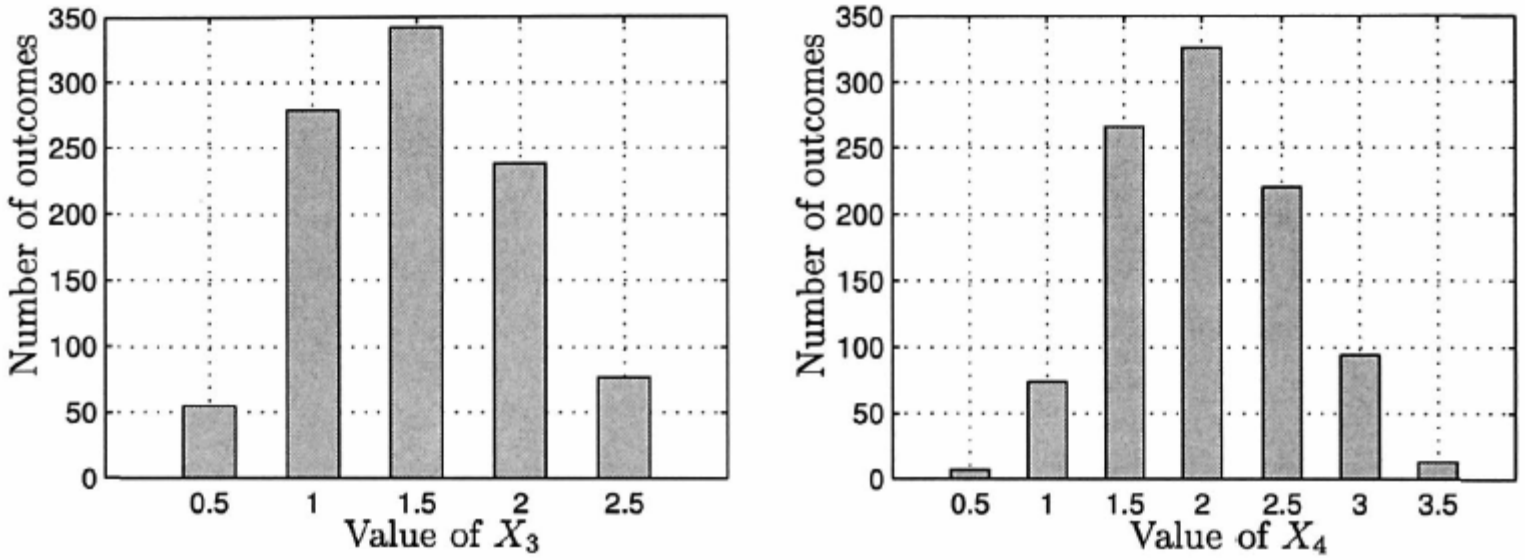
- Define the following variables

$$X_3 = U_1 + U_2 + U_3$$
$$X_4 = U_1 + U_2 + U_3 + U_4$$

- Then, we repeat the experiment several times and plot the corresponding histograms for X_3 and X_4.

Why Use Computer Simulation?

Histograms for addition of outcomes, $M = 1000$



Histograms appear more like a Guassian curve. Computer simulations indicate what we need to prove. However, they can not be used to prove theorems.

Figure source: S. M. Kay, *Intuitive Probabilty and Random Processes Using MATLAB*, Springer, 2006.

Computer Simulation of Random Experiments

Definition: A random variable X is the numerical outcome of an experiment.

Examples:

1. Number of dots in a die . X takes values in the set $\{1,2,3,4,5,6\}$ and is discrete.

2. Distance of dart from the center of a dartboard of radius 1. X takes values in the set $\{x: 0 \leq x \leq 1\}$ and is continuous.

Notation: We denote the random variable by a capital letter, say X , and its possible values by a small letter, say x_i for the discrete case and x for the continuous case.

Computer Simulation of Random Experiments

- A computer simulation can be used to determine several properties of X.
- We perform many experiments and observe the outcome for each experiment. The number of experiments will be denoted by M.
- To simulate a discrete random variable we use the MATLAB statement `rand`, which generates a number at random within the (0,1) interval.
- We provide a MATLAB program to generate 3 outcomes (realizations) of X whose possible values are $\{x_1, x_2, x_3\}$ with probabilities $\{p_1, p_{2,} p_3\}$.
- The program can easily be extended to generate N realizations of X whose possible values are $\{x_1, \dots, x_N\}$ with probabilities $\{p_1, \dots, p_N\}$ for arbitrary N.

Computer Simulation of Random Experiments

```
M = 1000;
x = zeros(M,1);
x1 = 3;
x2 = 7;
x3 = 9;
p1 = 0.2;
p2 = 0.5;
for i = 1:M i = 1:M
      u = rand(1,1);
      if u <= p1
         x(i,1) = x1;
      elseif u <= (p1+p2)
         x(i,1) = x2;
      else
         x(i,1) = x3;
      end
   end
```

Note: For a continuous random variable X that is Gaussian, the MATLAB statement `rand(M,1)` generates M realizations of X. Later, we will see how to generate realizations of other continuous random variables.

Determining Characteristics of Random Variables

- For a given continuous random variable, the probability per unit length is called the *probability density function* (PDF).
- Similary, for a given discrete random variable, the probability of being equal to a number is called the *probability mass function* (PMF).
- PDF or PMF estimation is one of the most important problems in probability.
- Once the PDF or PMF is estimated for the random variable of the interest, we can calculate quantities like

    i. Probability of an interval, ($P[a \leq X \leq b]$)
    ii. Average value
    iii. PDF or PMF of a function of a random variable.

Determining Characteristics of Random Variables

Example (PDF Estimation):

- Probabilities can be estimated by finding the histogram and then dividing the number of outcomes in each bin by the total number of realizations. Then, to estimate PDF, we divide the estimated probability by the bin width.

- To justify this, note that the probability of X taking on a value in an interval is the area under the PDF of that interval:

$$P[a \leq X \leq b] = \int_a^b p_X(x)dx$$

- Taking $a = x_0 - \Delta x/2,\ \ b = x_0 + \Delta x/2$ and assuming that Δx is small gives

$$P[x_0 - \Delta x/2 \leq X \leq x_0 + \Delta x/2] \approx p_X(x_0)\Delta x$$

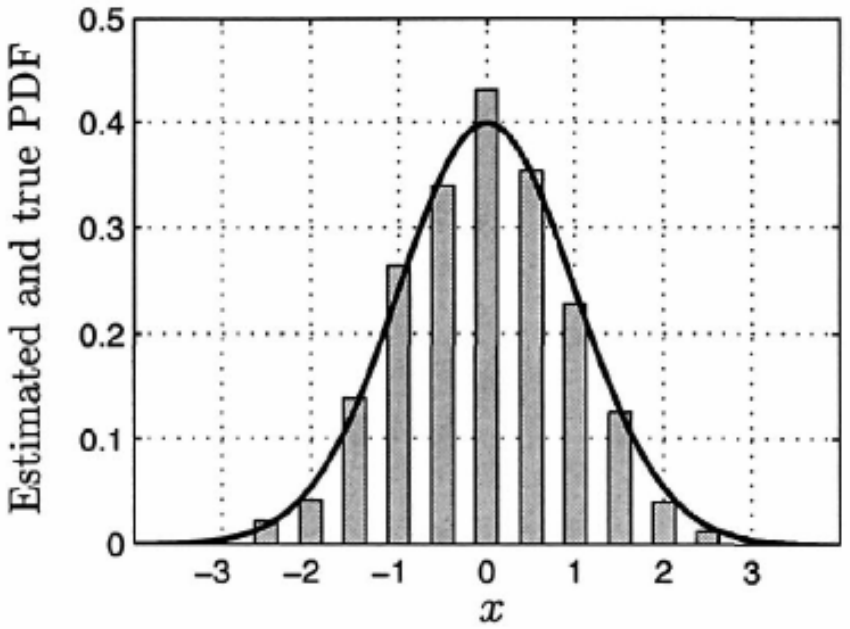
$$p_X(x_0) \approx \frac{P[x_0 - \Delta x/2 \leq X \leq x_0 + \Delta x/2]}{\Delta x}$$

Determining Characteristics of Random Variables

$M = 1000$

Bin centers: -3.5, -3, -2.5, -2, -1.5, -1, -0.5, 0, 0.5, 1, 1.5, 2, 2.5, 3, 3.5

$\Delta x = 0.5$



True PDF:

$$p_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

Estimated and true PDFs for a Gaussian random variable.

Figure source: S. M. Kay, *Intuitive Probabilty and Random Processes Using MATLAB*, Springer, 2006.

Determining Characteristics of Random Variables

Example (Probability of an interval):

- In the previous example, in order to estimate $P[a \leq X \leq b]$, M realizations of X are generated, the number of outcomes that fall into [a,b] interval is determined and divided by M, which should be large enough.

- For $a = 2$ and $b = \infty$, the true value is

$$P[X > 2] = \int_2^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx = 0.0228$$

- The results for an increasing number of M are given in the table.

M	Estimated $P[X > 2]$	True $P[X > 2]$
100	0.0100	0.0228
1000	0.0150	0.0228
10,000	0.0244	0.0288
100,000	0.0231	0.0288

Table source: S. M. Kay, *Intuitive Probabilty and Random Processes Using MATLAB*, Springer, 2006.

Determining Characteristics of Random Variables

Example (Average Value):

- In the previous example, in order to estimate the average (expected or mean) value of X, M realizations of X are generated and the following quantitiy is calculated

$$\frac{1}{M}\sum_{i=1}^{M} x_i$$

- As we will see, the true mean is 0. Estimated means for different values of M are given in the table.

M	Estimated mean	True mean
100	0.0479	0
1000	−0.0431	0
10,000	0.0011	0
100,000	0.0032	0

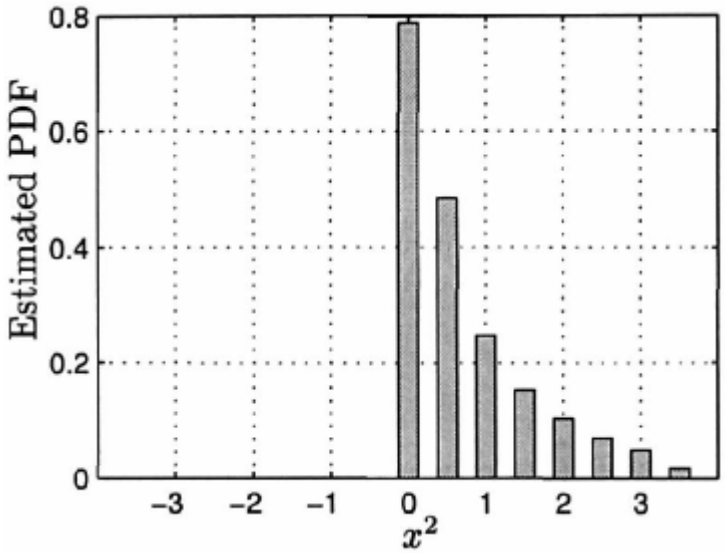
Table source: S. M. Kay, *Intuitive Probabilty and Random Processes Using MATLAB*, Springer, 2006.

Determining Characteristics of Random Variables

Example (Transformation of a Random Variable):

- Let us define $Y = X^2$. Characteristics of Y can be estimated from the realizations of X.

- The estimated PDF for M = 1000 and the the estimated means for different values of M are given below.



M	Estimated mean	True mean
100	0.7491	1
1000	0.8911	1
10,000	1.0022	1
100,000	1.0073	1

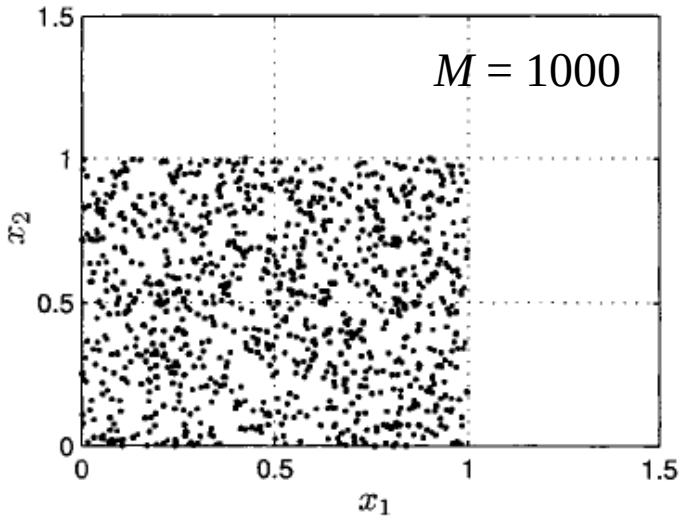
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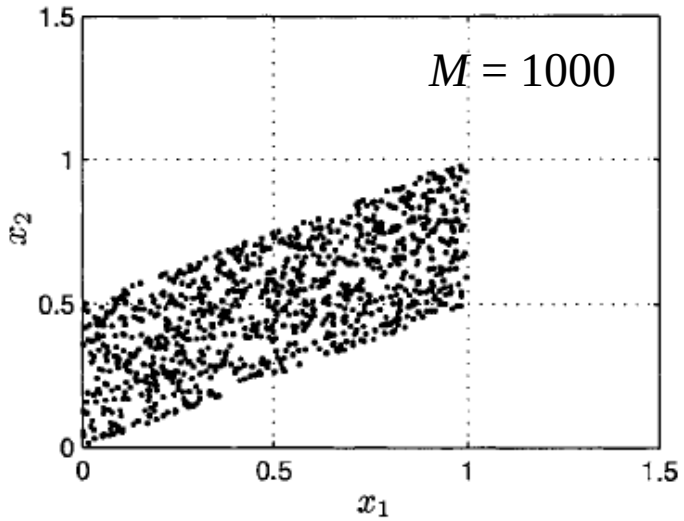
Determining Characteristics of Random Variables

Example (Multiple Random Variables):

- Let us choose a point in the square $\{(x, y): 0 \leq x \leq 1, 0 \leq y \leq 1\}$ according to a rule. This experiment produces a random vector $[X_1, X_2]^T$.

- The rule may or may not cause the value of x_2 to depend on the value of x_1.





- Can you say whether X_1 and X_2 are independent from the scatter plots? Why?

Figure source: S. M. Kay, *Intuitive Probabilty and Random Processes Using MATLAB*, Springer, 2006.

Real World Example: Digital Communication

- In a digital communication system, 0 and 1 bits are communicated to a receiver by sending signals $s_0(t)$ and $s_1(t)$, where

$$s_0(t) = A\cos(2\pi F_0 t + \pi), \qquad s_1(t) = A\cos(2\pi F_0 t)$$

- The receiver that is used to decode the transmitted bit is given by

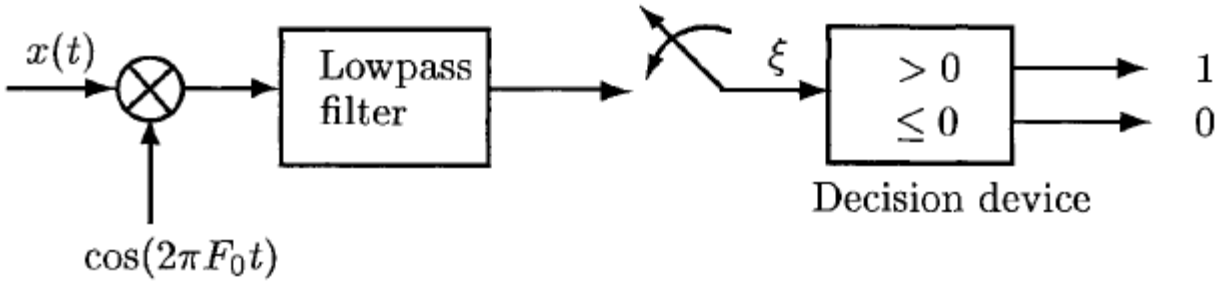


Figure source: S. M. Kay, *Intuitive Probabilty and Random Processes Using MATLAB*, Springer, 2006.

Real World Example: Digital Communication

- The output of the multipler is given by

$$= \begin{cases} s_0(t)\cos(2\pi F_0 t) = -A\left[\frac{1}{2} + \frac{1}{2}\cos(4\pi F_0 t)\right], & \text{for 0} \\ s_1(t)\cos(2\pi F_0 t) = A\left[\frac{1}{2} + \frac{1}{2}\cos(4\pi F_0 t)\right], & \text{for 1} \end{cases}$$

- After the low pass filter, which filters the high frequency part ($\cos(4\pi F_0 t)$) of the signal, and sampler we have

$$\Delta = \begin{cases} -A/2, & \text{for 0} \\ A/2, & \text{for 1} \end{cases}$$

- The receiver decides a 1 was transmitted if $\Delta > 0$, otherwise it decides a 0 was transmitted.

Real World Example: Digital Communication

- In reality, the transmitted signal is not equal to the signal receiver gets because of the channel noise. The actual value of Δ observed (W represensts noise) is

$$\Delta = \begin{cases} -A/2 + W, & \text{for 0} \\ \mathrm{A}/2+\mathrm{W}, & \text{for 1} \end{cases}$$

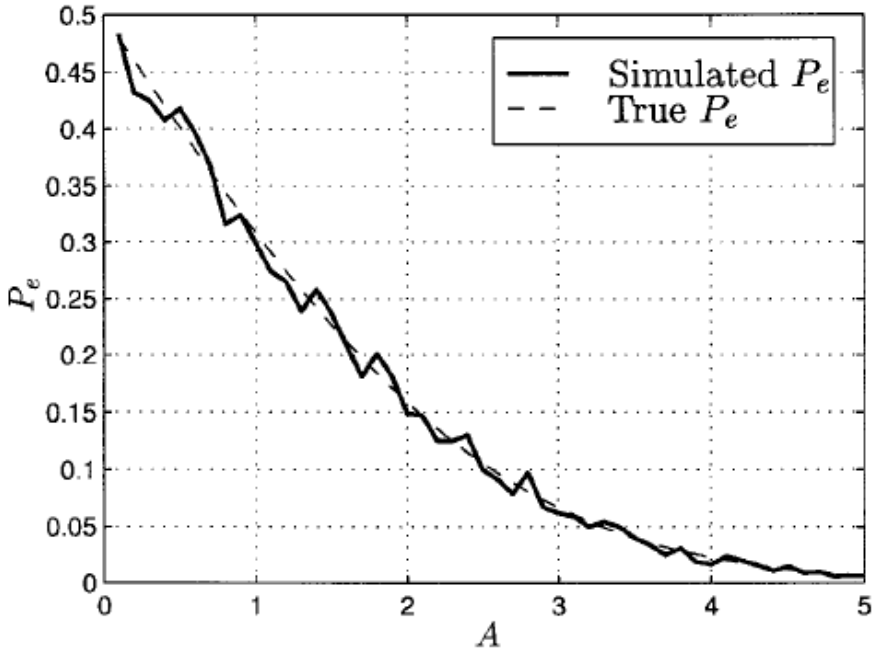
- Consider the case of 1 having been transmitted. The receiver makes an error if $A/2+W \leq 0$. The chance of this called the *probability of error* and denoted by P_e. That is

$$P_e = P[A/2 + W \leq 0]$$

- The error depends on the signal amplitude A (why and how?)

Real World Example: Digital Communication

Using a computer simulation, we can plot P_e as a function of A. Assuming noise is Gaussian gives



P_e decreases as the signal amplitude increases. What is the physical interpretation?

Figure source: S. M. Kay, *Intuitive Probabilty and Random Processes Using MATLAB*, Springer, 2006.